

Project Initialization and Planning Phase

Date	16 August 2026
Team ID	xxxxxxx
Project Title	India's Agricultural crop production Analysis
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	Visualize Trends: Identify multi decades crop output cycle, seasonal variations , and state level productivity. • Empower Stakeholders: Provide prescriptive guidance for farmers, policymakers, and agribusiness investors. Build Interactive Dashboards: Construct intuitive Tableau stories, map overlays, and drill down interfaces. Optimize Yields: Highlight top-performing crops and isolate low-yield zones for focus ed intervention.
Scope	• Data Preparation: Ingestion, cleaning, and standardization of 1997–2021 agricultural data from open repositories. Geographical & Seasonal Coverage: Spans 30+ states, Union Territories, and all cropping seasons (Kharif, Rabi, Summer, Autumn, Whole Year Visual Engineering: Development of dynamic Tableau worksheets, KPI metrics, geospatialmaps, and user filters Performance & Documentation: Optimized data extraction.
Problem Statement	
Description	? Analyzes two decades of Indian agricultural data (1997–2021) to map regional productivity and crop performance. ? Cleans and transforms complex multi-year agricultural datasets into standardized analytical formats.

	☑ Leverages interactive Tableau dashboards to visually track seasonal yield fluctuations and regional disparities. ☑ Evaluates trends across diverse cropping seasons (Kharif, Rabi, Whole Year) and major Indian states.
Impact	For Farmers: Reduces financial and climatic vulnerability via validated crop-yield insights. • For Policymakers: Facilitates targeted agricultural credit, irrigation planning,. • For Agribusinesses: Decreases market uncertainty through reliable crop supply forecast
Proposed Solution	
Approach	• Data Ingestion & Cleaning: Aggregate multi-decade raw datasets (1997–2021), clean duplicate/null records, and normalize key variables such as production quantity, cultivated area, and yield metrics. • Exploratory Analytics: Perform statistical analysis to categorize regional performance, identify dominant crop varieties, and isolate long-term trends across different agricultural zones. • Visual Data Architecture: Build custom dynamic calculations and hierarchical data structures within Tableau to support cross-filtering between states, districts, seasons, and years. • Dashboard Design & Deployment: Construct specialized interactive stories and drill-down visual interfaces tailored to farmers, policy analysts, and agribusiness investors.
Key Features	• Geospatial Yield Mapping: Interactive state and district maps displaying crop density, yield distribution, and regional production highlights. • Multi-Year Trend Analysis: Line and area charts tracking production volume and land usage over 24+ years to map growth trajectories. • Season-Wise Drill Downs: Comparative visual matrices isolating Kharif, Rabi, Summer, and Whole Year cropping patterns. Interactive Filtering: Cross-functional KPI cards, drop-down filters, and top N state/crop visual summaries for dynamic stakeholder exploration

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	e.g., 2 x NVIDIA V100 GPUs
Memory	RAM specifications	e.g., 8 GB
Storage	Disk space for data, models, and logs	e.g., 1 TB SSD
Software		
Frameworks	Python frameworks	e.g., Flask
Libraries	Additional libraries	e.g., scikit-learn, pandas, numpy
Development Environment	IDE, version control	e.g., Jupyter Notebook, Git
Data		
Data	Source, size, format	e.g., Kaggle dataset, 10,000 images